

JavaScript Complete Course (Basic to Advanced)

Module 1: Introduction to JavaScript

Lesson 1

- What is JavaScript?
- History of JavaScript
- ECMAScript (ES6+)
- JavaScript vs Java
- Client-side vs Server-side JavaScript
- Features of JavaScript

First JavaScript Program

```
console.log("Hello World");
```

Lesson 2

Adding JavaScript to HTML

- Inline JavaScript
- Internal JavaScript
- External JavaScript

```
<script>  
  alert("Welcome");  
</script>  
<script src="script.js"></script>
```

Module 2: Variables & Data Types

Lesson 3

Variables

- var
- let
- const

Data Types

- Number
- String
- Boolean
- Undefined
- Null
- BigInt
- Symbol

Example

```
let name = "Anurag";  
const age = 22;
```

Module 3: Operators

Lesson 4

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Increment & Decrement
- Ternary Operator

Practice

Simple Calculator

Module 4: Input & Output

Lesson 5

- alert()
- prompt()
- confirm()
- console.log()
- document.write()

Project

Student Marks Calculator

Module 5: Conditional Statements

Lesson 6

- if
- if...else
- else if
- switch

Projects

- Grade Calculator
 - Voting Eligibility
 - Electricity Bill
-

Module 6: Loops

Lesson 7

- for
- while
- do...while
- break
- continue

Projects

- Multiplication Table
- Prime Numbers
- Factorial

Module 7: Functions

Lesson 8

- Function Declaration
- Function Expression
- Arrow Functions
- Parameters
- Return Values
- Default Parameters

Projects

- EMI Calculator
 - BMI Calculator
-

Module 8: Strings

Lesson 9

Methods

- length
- slice()
- substring()
- replace()
- split()
- trim()
- includes()
- startsWith()
- endsWith()

Projects

- Password Validator
 - Email Checker
-

Module 9: Arrays

Lesson 10

- Creating Arrays
- Accessing Elements
- Updating Arrays

Methods

- push()
- pop()
- shift()
- unshift()
- splice()
- slice()

Projects

- Student Marks
 - Shopping Cart
-

Module 10: Objects

Lesson 11

- Object Creation
- Properties
- Methods
- Nested Objects

Projects

- Employee Database
 - Student Information
-

Module 11: Array Methods

Lesson 12

Methods

- forEach()
- map()
- filter()
- reduce()
- find()
- some()
- every()
- sort()

Projects

- Student Results
 - Employee Salary
-

Module 12: DOM Introduction

Lesson 13

DOM

- getElementById()
- getElementsByClassName()
- getElementsByTagName()
- querySelector()
- querySelectorAll()

Projects

- Change Text
 - Change Images
-

Module 13: DOM Manipulation

Lesson 14

- innerHTML
- innerText
- textContent

- createElement()
- append()
- remove()
- replaceChild()

Projects

- Todo List
 - Notes App
-

Module 14: CSS Manipulation

Lesson 15

- style
- classList
- add()
- remove()
- toggle()

Projects

- Dark Mode
 - Theme Switcher
-

Module 15: Events

Lesson 16

Events

- click
- dblclick
- mouseover
- mouseout
- keyup
- keydown
- input
- submit
- change

Projects

- Image Slider
 - Keyboard Event Viewer
-

Module 16: Forms

Lesson 17

- Form Validation
- Required Fields
- Regular Expressions
- Password Validation

Projects

- Registration Form
 - Login Form
-

Module 17: Date & Time

Lesson 18

Date Object

Methods

- getDate()
- getMonth()
- getFullYear()
- getHours()
- getMinutes()

Projects

- Digital Clock
 - Age Calculator
-

Module 18: Math Object

Lesson 19

Methods

- `round()`
- `ceil()`
- `floor()`
- `random()`
- `max()`
- `min()`

Projects

- Dice Game
 - Number Guessing Game
-

Module 19: Timers

Lesson 20

- `setTimeout()`
- `setInterval()`
- `clearTimeout()`
- `clearInterval()`

Projects

- Stopwatch
 - Countdown Timer
-

Module 20: ES6 Features

Lesson 21

- Template Literals
- Destructuring

- Spread Operator
- Rest Parameters
- Default Parameters
- Optional Chaining

Projects

- Employee Management
-

Module 21: Advanced Functions

Lesson 22

- Callback Functions
- Higher Order Functions
- Closures
- Lexical Scope

Projects

- Custom Calculator
-

Module 22: Error Handling

Lesson 23

- try
- catch
- finally
- throw

Projects

- Form Error Handler
-

Module 23: JSON

Lesson 24

- JSON.parse()
- JSON.stringify()

Projects

- Student Records
-

Module 24: Local Storage

Lesson 25

- localStorage
- sessionStorage
- Cookies (Introduction)

Projects

- Todo App
 - Notes App
-

Module 25: Asynchronous JavaScript

Lesson 26

- Synchronous vs Asynchronous
- Callback
- Promise
- Async/Await

Projects

- Delayed Message
 - Fake API Simulation
-

Module 26: Fetch API

Lesson 27

- Fetch()
- GET
- POST
- JSON Data

Projects

- Weather App
 - User Details Viewer
-

Module 27: Modules

Lesson 28

- import
- export
- Module Organization

Projects

- Calculator Modules
-

Module 28: Object-Oriented JavaScript

Lesson 29

- Classes
- Objects
- Constructor
- Inheritance
- Static Methods
- Encapsulation

Projects

- Bank Management System
 - Library Management
-

Module 29: Modern JavaScript

Lesson 30

Topics

- Optional Chaining
 - Nullish Coalescing
 - Dynamic Imports
 - ES2023 Features (Overview)
 - Best Practices
 - Code Optimization
-

Module 30: Final Projects

Build Complete Real-World Applications

Projects

1. Calculator
2. Digital Clock
3. Stopwatch
4. Countdown Timer
5. BMI Calculator
6. Age Calculator
7. Student Result System
8. Employee Management
9. Quiz App
10. Weather App
11. Currency Converter
12. Todo App
13. Notes App
14. Expense Tracker
15. Shopping Cart
16. E-commerce Product Page
17. Login System
18. Registration System
19. Portfolio Website

20. Responsive Company Website
 21. Movie Search App
 22. Music Player
 23. Image Slider
 24. Form Validator
 25. Recipe Finder
 26. Random Quote Generator
 27. Password Generator
 28. Dashboard UI
 29. Chat Application (Frontend)
 30. Mini E-commerce Website
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30 Hands-on Practice Programs

1. Hello World
2. Even or Odd Number
3. Prime Number
4. Factorial
5. Fibonacci Series
6. Reverse a String
7. Palindrome Checker
8. Largest Number
9. Armstrong Number
10. Calculator
11. Student Grade Calculator
12. Multiplication Table
13. Guess the Number
14. Dice Roll Simulator
15. Random Password Generator
16. Temperature Converter
17. Currency Converter
18. Digital Clock
19. Stopwatch
20. Countdown Timer
21. Todo List
22. Notes App
23. Quiz Application
24. Weather Application
25. Expense Tracker
26. Student Management System
27. Employee Payroll System
28. Shopping Cart
29. Movie Search App
30. Portfolio Website

JavaScript Interview Questions (Basic to Advanced)

Beginner

1. What is JavaScript?
2. Difference between `var`, `let`, and `const`?
3. What are JavaScript data types?
4. Explain `null` and `undefined`.
5. What are operators?
6. Difference between `==` and `===`?
7. What are functions?
8. What is an array?
9. What is an object?
10. Explain loops in JavaScript.

Intermediate

11. What is the DOM?
12. Difference between `innerHTML` and `innerText`?
13. Explain event bubbling and capturing.
14. What is event delegation?
15. Explain callbacks.
16. What are promises?
17. What is `async/await`?
18. What is JSON?
19. Explain `localStorage` and `sessionStorage`.
20. What are arrow functions?

Advanced

21. Explain closures.
22. What is lexical scope?
23. What are higher-order functions?
24. Explain hoisting.
25. What is the event loop?
26. Explain the call stack.
27. What is the difference between synchronous and asynchronous JavaScript?
28. Explain prototypes and inheritance.
29. What are ES6 modules?
30. What are JavaScript best practices?

